SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code: CSA17 | Course: Artificial Intelligence | CO1 – Apply (BL3)

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Answer Sheet – Analytical Problem Solving

The solutions below follow the assessment requirement to state the problem interpretation, select and justify the method, show the solution process and calculations, and interpret the final result.

Question 1: Two-Room Vacuum World

Problem Interpretation: The vacuum agent starts in square A; both A and B are dirty. A clean square contributes +1 at the end of every time step. Every Left/Right movement incurs penalty p. The horizon is T = 6 steps.

Method Selection: Trace each agent step-by-step using its stated action rule. For every step, compute clean-square reward minus movement penalty. Then express the total score algebraically in p.

(a) PEAS Description

Component	Description
Performance	+1 for every clean square at the end of each time step; –p for every Left/Right move.
Environment	Two-square vacuum world: A and B; initially both dirty; cleaned dirt does not reappear.
Actuators	Suck, Right, Left, NoOp.
Sensors	Perceives its current location and whether the current square is Dirty or Clean.

(b) Agent X – Simple Reflex Trace

Step	Action	Clean squares at end	Move penalty (p=1)	Step score
1	Suck	1	0	1
2	Right	1	1	0
3	Suck	2	0	2
4	Left	2	1	1
5	Right	2	1	1
6	Left	2	1	1
	Total		**3**	**6**

(c) Agent Y – Reflex with Internal State

Step	Action	Clean squares at end	Move penalty (p=1)	Step score
1	Suck	1	0	1
2	Right	1	1	0
3	Suck	2	0	2
4	NoOp	2	0	2
5	NoOp	2	0	2
6	NoOp	2	0	2
	Total		**1**	**9**

(d) General score as a function of p

Agent X: reward from clean squares = 1 + 1 + 2 + 2 + 2 + 2 = 10. It makes 3 moves, so $\mathbf{S_X(p) = 10 - 3p}$.

Agent Y: reward from clean squares = 10. It makes only 1 move, so $\mathbf{S_Y(p) = 10 - p}$.

Y is strictly better when 10 – p > 10 – 3p, i.e. **2p > 0**, so **p > 0**. At p = 0, both have the same score. (If negative penalties are mathematically allowed, the inequality is p > 0.)

(e) Result Interpretation

Agent Y is rational under the stated performance measure because, after both squares are known to be clean, its internal state allows it to choose NoOp and avoid unnecessary movement penalties. Agent X reacts only to the current percept. Once it reaches a clean square, it cannot remember that the other square has already been cleaned, so it keeps moving. Thus, identical current percepts can require different rational actions when percept history contains useful information.

Question 2: Task Environment Analysis and Agent Selection

Problem Interpretation: Classify E1–E3 on observability, number of agents, determinism, episodic/sequential, static/dynamic, and discrete/continuous. Then score each property using the stated difficulty convention.

Method Selection: Assign 1 point to each harder property and 0 to each easier property; assign 0.5 for semi-dynamic. Sum the six scores and rank the environments.

Property	E1: Autonomous taxi	E2: Part-picking robot	E3: Chess agent
Observability	Partially observable – traffic state is not fully known.	Partially observable – camera may not reveal all defect/motion information.	Fully observable – board state is visible.
Agents	Multi-agent – other road users act independently.	Single agent – robot is the decision-making agent.	Multi-agent – opponent is an active agent.
Determinism	Stochastic – traffic events are uncertain.	Stochastic – defects and conveyor motion introduce uncertainty.	Deterministic – legal moves have defined outcomes.
Episodic / Sequential	Sequential – earlier actions affect later driving.	Sequential – picking actions occur over a continuing process.	Sequential – earlier moves affect later positions.
Static / Dynamic	Dynamic – traffic changes while deciding.	Dynamic – conveyor is moving.	Semi-dynamic – board is static, but the clock changes.
Discrete / Continuous	Continuous – position, speed and time vary continuously.	Continuous – object position/motion vary continuously.	Discrete – finite board states and legal moves.

(b) Difficulty Scores

Environment	O	A	D	S	St/Dy	Disc/Cont	Total
E1 – Taxi	1	1	1	1	1	1	**6.0**
E2 – Robot	1	0	1	1	1	1	**5.0**
E3 – Chess	0	1	0	1	0.5	0	**2.5**

Ranking (hardest → easiest): E1 (6.0) > E2 (5.0) > E3 (2.5).

(c) Agent Architecture Selection

Environment	Selected architecture	Justification
Hardest: E1 Taxi	Learning / Utility-based	The taxi must handle uncertainty, changing traffic, multiple agents and continuous decisions. A utility-based component can trade off safety, time and comfort, while learning can improve performance from experience.
Easiest: E3 Chess	Utility-based	Chess is fully observable and discrete, but it is sequential and multi-agent. A utility-based agent can evaluate board positions and select moves that maximize expected game outcome.

(d) Result Interpretation: Domain name alone does not determine agent complexity. The environment properties determine what the agent must handle. Partial observability requires inference; stochasticity requires uncertainty handling; sequential and dynamic environments require memory and planning; continuous environments require richer control. Therefore E1 needs a more complex agent than the relatively structured E3 environment.

Question 3: Turing Test Data Across the History of AI

Problem Interpretation: Each system faced 50 interrogations while being the machine. Correct identification means the judge correctly recognized the machine; fooling means the machine was taken for a human.

Method Selection: Correct-identification rate = correctly identified ÷ 50 × 100. Fooling rate = (50 – correctly identified) ÷ 50 × 100. Compare correct-identification rates with Turing's 70% reference and with the 50% guessing level.

System	Correctly identified	Correct-identification rate	Fooling count	Fooling rate
ELIZA	47/50	94%	3/50	6%
Chatbot B	36/50	72%	14/50	28%
System C	29/50	58%	21/50	42%

(b) Turing 70% criterion and distance from chance

System	Correct ID rate	Satisfies ≤70% criterion?	Distance from 50% chance
ELIZA	94%	No	44 percentage points
Chatbot B	72%	No	22 percentage points
System C	58%	Yes	8 percentage points

Method: use the correct-identification rate directly. The reference says an average interrogator would have no more than a 70% chance of making the right identification. System C is closest to the 50% guessing level among the three.

(c) Improvement in fooling rate

ELIZA → System C: 42% – 6% = **36 percentage points**. Ratio = 42 ÷ 6 = **7.0×** (System C's fooling rate is seven times ELIZA's).

Chatbot B → System C: 42% – 28% = **14 percentage points**.

(d) Result Interpretation

The Turing Test measures whether a system can *act humanly* in a conversational setting. A high fooling rate does not establish that the system is thinking or acting rationally. The result only shows that judges sometimes cannot distinguish the machine from a human under the test conditions. Therefore System C's 42% fooling rate is evidence about human-like conversational behaviour, not proof of genuine understanding, consciousness, or rationality.

Question 4: Rational Route Choice under Uncertainty

Problem Interpretation: The taxi must reach the airport within a 55-minute deadline. Utility is U = 100 – T – 20L, where L = 1 for a late arrival and 0 otherwise. The rational route is the route with the highest expected utility based on available percept information.

Method Selection: Calculate utility separately for normal and disrupted outcomes. Then calculate expected utility as Σ(probability × outcome utility). Choose the route with maximum expected utility.

Route	Outcome	Time T	Late L?	Utility U
R1	Normal	30 min	No (0)	100–30 = **70**
R1	Disrupted	70 min	Yes (1)	100–70–20 = **10**
R2	Normal	45 min	No (0)	100–45 = **55**
R2	Disrupted	60 min	Yes (1)	100–60–20 = **20**
R3	Normal	50 min	No (0)	100–50 = **50**

(b) Expected utility

Route	Expected utility calculation	Expected utility
R1	0.70(70) + 0.30(10)	**52.0**
R2	0.90(55) + 0.10(20)	**51.5**
R3	1.00(50)	**50.0**

Decision: Choose **R1**, because 52.0 is the highest expected utility.

(c) New percept: highway disruption probability rises to 0.80

New EU(R1) = 0.20(70) + 0.80(10) = 14 + 8 = **22**.

The other expected utilities remain R2 = 51.5 and R3 = 50.0. Therefore the new rational choice is **R3 (Ring road)**, because 50.0 is now the highest expected utility.

(d) Actual outcome and rationality

If the agent selected R1 under the original information and the disruption occurred, the actual utility is **10**. The agent was **not irrational**: at the time of decision it selected R1 because its expected utility (52.0) was higher than R2 (51.5) and R3 (50.0).

A **rational agent** chooses the action that maximizes expected performance based on its percept sequence and available knowledge. An **omniscient agent** would know the actual future outcome in advance and could choose the best route for that realized situation. Rationality does not require omniscience.

(e) Result Interpretation

Rationality depends on both the percept sequence and the performance measure. The radio alert changes the agent's information about the probability of a highway accident, so the expected utility of R1 falls from 52.0 to 22.0. The physical roads did not change; the agent's information changed, and therefore its rational decision changed.

Final Answer Summary

Question	Key final result
Q1	PEAS identified; X score = 6 at p=1; Y score = 9 at p=1; Sx=10–3p, Sy=10–p; Y better for p>0; Y is rational under the given measure.
Q2	Difficulty: E1=6.0, E2=5.0, E3=2.5; ranking E1 > E2 > E3.
Q3	Correct ID: ELIZA 94%, B 72%, C 58%; fooling: 6%, 28%, 42%; only C satisfies ≤70% criterion.
Q4	Original EU: R1=52.0, R2=51.5, R3=50.0 → R1. After highway risk rises to 0.80: EU(R1)=22 → R3.

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